

Lesson 17 Relations

1. Let $A = \{0, 1, 2, 3, 4\}$. For each relation R on A , write out the set R and give a diagram.
 - a) Let R be the “ $\geq$ ” relation.
 - b) Let R be the “divides” relation.
 - c) Let $R = \{(x, y) \in A \times A : x - y = 1\}$
 - d) Let $R = \{(x, y) \in A \times A : 2 \mid (x - y)\}$
2. For each of the relation R in Question 1, determine if R reflexive, symmetric, or transitive.
3. For each of the relation R in Question 1, write R^{-1} in set-builder notation.
4. Let R be a relation and B be a set. Then $R \cap (B \times B)$ is the **restriction** of R to B . For the relations in Question 1, find the restriction of R to $B = \{1, 2, 3\}$.
5. Let $A = \{1, 2\}$ and R be the subset relation on $\mathcal{P}(A)$. List the elements of R .
6. Let A be a set with n elements. How many relations on A are possible?
7. Let R be a relation. Prove that $(R^{-1})^{-1} = R$.
8. Let R be a relation. Prove that $R = R^{-1}$ if and only if R is symmetric.